



Research Article

Multi-strategy combined bionic coating for long-term robust protection against marine biofouling

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ABSTRACT

Polyurethane-fluorinated polysiloxane (PU-^FPDMS) with high-strength, high-bonding and low surface energy is synthesized as the matrix, and the PU-^FPDMS/MCs/Ag marine anti-fouling coating on the surface of imitation crab shells is constructed by assembling butenolide@1,1-stilbene-modified hydrolyzed polyglycidyl methacrylate/graphene oxide microcapsules (Bu@PGMA_m/GO MCs) with compact multi-shell structure and Ag nanoparticles (AgNPs) step by step on the PU-^FPDMS matrix. The PU-^FPDMS/MCs/Ag bionic anti-fouling coatings achieve long-term and stable anti-fouling effect under the combination of robust low-surface-energy PU-^FPDMS matrix, steady-state sustained release of butenolide encapsulated by the compact multi-shell, bionic surface formed by the microcapsules and AgNPs, and the release of Ag⁺. The shear strength, tensile strength, and elongation at break of the PU-^FPDMS/MCs/Ag are 3.53 MPa, 6.7 MPa, and 192.83 %, respectively. Its static contact angle and sliding angle are 161.8° and 3.6°, respectively. The antibacterial rate of PU-^FPDMS/MCs/Ag against *Escherichia coli*, *Staphylococcus aureus*, and *Candida albicans* can reach 100 %. Compared with glass blank, PU, PU-^FPDMS, PU-^FPDMS/Ag, and PU-^FPDMS/MCs, both the adhesion number and coverage percentage of *Chlorella* adhere to PU-^FPDMS/MCs/Ag are the minimum values, which are 600 cell mm⁻² and 1.53 %, respectively. After 6 months of marine field test, the primer blank, PU, PU-^FPDMS all show different degrees of attachment by shellfish, spirorbis, algae and other biofouling, while the PU-^FPDMS/MCs/Ag coating is still not covered with biofouling, while the PU-^FPDMS/MCs/Ag coatings still exhibit little attachment of marine fouling. The PU-^FPDMS/MCs/Ag bionic anti-fouling coatings are expected to be widely used in the fields of anti-fouling, anti-icing, anti-fogging, drag reduction, self-cleaning, and antibacterial.

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1. Introduction

Ships, large machinery, and other structures in the seawater environments are prone to biofouling due to the attachment of numerous marine organisms [1,2]. Marine fouling organisms cause significant harm to ships and marine facilities by increasing fuel consumption, interfering with the normal operation of sonar systems, and accelerating corrosion and damage. In order to prevent marine biofouling, the most effective, economical, and comprehensive approach is to apply anti-fouling coatings to the surface of marine structures [3,4]. At present, there are three main types

of marine anti-fouling coatings. First, agent-releasing anti-fouling coatings rely on hydrolysis and degradation of the polymer matrix to obtain anti-fouling agent release. Second, low-surface energy anti-fouling coatings rely on low-surface energy polymers to obtain the desorption of fouling organisms under medium and high-speed water flow. Third, bionic surface anti-fouling coatings rely on surface microstructure to reduce the contact points of fouling organisms.

The anti-fouling agent-releasing anti-fouling coatings are made by mixing resins such as acrylates, vinyl resins, polyurethanes, and epoxy resins, as well as anti-fouling agents, additives, and other components, and then forming after coating. Since the late 1950s, tributyltin anti-fouling agents have been widely used due to their excellent anti-fouling effect [5]. However, its accumulation in seawater can lead to genetic variation in marine organisms, which was banned in 2008. With attention to marine biosecurity,

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researchers have begun to extract a large number of natural anti-fouling agents with anti-fouling activity from marine bacteria, algae, invertebrates, and terrestrial plants. These natural antifouling agents contain more than 60 kinds of natural compounds such as organic acids, inorganic acids, lactones, terpenes, phenols, indoles, camptothecin, capsaicin, and piperine [6,7]. For example, researchers [8] obtained a new class of marine natural product anti-fouling agent 5-octylfuran-2(5H)-one (butenolide, named as Bu) by modifying the chemical structure of metabolites of marine streptomyces. Different from the traditional anti-fouling agent, Bu achieves the purpose of preventing biofouling by blocking the signals and metabolic pathways related to the attachment of fouling organisms. It has high efficiency and broad-spectrum anti-fouling activity for hard-shell and soft-bodied organisms. Moreover, it is easy to degrade without environmental accumulation risk and has a simple structure, easy to synthesize, low production cost, and good stability in the polymer matrix, making it the first choice for green and environmentally friendly anti-fouling agents [9–11]. At present, the researches on Bu only focus on its synthesis methodology [12–14], its interference with marine larval sedimentation [15–17], and its toxicology of marine organisms in the laboratory [18–20], while the research on its application in marine anti-fouling coatings and actual marine anti-fouling need to be strengthened. Low-surface-energy anti-fouling coating is a kind of anti-fouling coating that is obtained by using low-surface-energy polymer of organofluorine and organosilicon as matrix resins. The adhesion between this kind of coating and fouling organisms is weak, and it can be separated by water washing or mechanical cleaning [21]. However, the anti-fouling effect of the low-surface-energy coating is not obvious for static conditions, and its mechanical strength is poor, so it is easy to be damaged in the mechanical cleaning of fouling organisms. In addition, the adhesion of the low-surface-energy coating to the substrate is weak and easy to peel off. Therefore, scholars have carried out a lot of research on improving the mechanical properties and bonding properties of low-surface-energy anti-fouling coating. For example, the incorporation of nanoparticles (carbon nanotubes, graphene, natural sepiolite, etc.) can improve the mechanical strength of the coating [22–28]. Furthermore, polyurethane and polyurea segments are incorporated into the molecular structure of the coating. The bonding strength is enhanced by the formation of hydrogen bonds between the active groups on the substrate and urethane or urea bonds in the coating [29,30]. Inspired by the idea that the surface of marine animals (sharks, whales, dolphins, crabs, etc.) should not adhere to fouling organisms, the bionic anti-fouling coating achieves the purpose of reducing the adhesion contact point by obtaining a microscopic structure as regular as the skin structures of marine animals [31,32]. Although there are many methods for constructing microstructured surfaces, physical methods such as lithography, laser ablation, and molding, as well as chemical synthesis methods. However, it is still difficult to construct large-scale homogeneous microstructure surfaces at a low cost, and it remains to be verified whether these microstructures can be maintained in the marine environment for a long time and achieve long-term anti-fouling [33].

In order to solve the problems of anti-fouling agent-releasing type, low-surface-energy type, and bionic surface type anti-fouling coating, a robust low-surface-energy polyurethane-fluorinated polysiloxane (PU-^FPDMS) was synthesized in this work to obtain an effective balance of surface energy, mechanical strength, and bonding strength. On the basis of the research of microcapsules in our research group [34], Bu@PGMA_m/GO microcapsules with compact multi-shell structures were synthesized, and Bu anti-fouling agent was encapsulated by the microcapsules to achieve steady-state sustained release. By manipulating the step-by-step assembly of microcapsules and Ag nanoparticles in the PU-^FPDMS

matrix, a biomimetic “papillae” structure similar to the surface of crab shells was constructed to reduce the contact points of fouling organisms. Microcapsules (MCs) and Ag nanoparticles (AgNPs) were assembled in the PU-^FPDMS matrix in steps to construct a bionic “mastoid” structure similar to the surface of the crab shell and to reduce the contact point of fouling organisms. The release of broad-spectrum and high-efficiency green anti-fouling agents Bu and AgNPs plays a role in blocking and driving away marine organisms. Under the combined action of the above multiple strategies, the purpose of environmentally friendly, broad-spectrum, and long-term anti-fouling of PU-^FPDMS/MCs/Ag bionic anti-fouling coating was obtained. Among them, the multi-strategy combined anti-fouling mechanism is shown in Fig. 1.

The chemical composition, structure, stability and release behavior of Bu@PGMA_m/GO microcapsules are characterized by optical microscopy (OM), scanning electron microscopy (SEM), particle size distribution instrument, Fourier transform infrared spectroscopy (FT-IR), nuclear magnetic resonance spectroscopy (NMR), X-ray diffraction spectroscopy (XRD), Raman spectroscopy, X-ray photoelectron spectroscopy (XPS) and thermogravimetric analysis (TGA), and ultraviolet-visible spectrophotometry. The effects of a multi-strategy approach for steady-state sustained release of Bu, robust low-surface-energy matrix, bionic surface, and Ag release on the mechanical properties, self-cleaning performance, electrochemical performance, anti-bacterial performance, anti-algae attachment performance, and resistance to adhesion of marine organisms of PU-^FPDMS/MCs/Ag bionic anti-fouling coatings are investigated by universal testing machine, contact angle measuring instrument, electrochemical workstation, SEM, atomic force microscope (AFM) and by anti-bacterial test, anti-algae attachment test, and marine field test.

2. Experimental section

2.1. Materials

Phosphorus pentoxide (P₂O₅, 98 %), potassium persulfate (K₂S₂O₈, 98 %), sulfuric acid (H₂SO₄, 98 %), hydrochloric acid (HCl, 37 %), hydrogen peroxide (H₂O₂, 30 %), potassium permanganate (KMnO₄, 99 %), glycidyl methacrylate (GMA, 95 %), 1,1-stilbene (DPE, 97 %), divinylbenzene (DVB, 80 %), polyvinylpyrrolidone (PVP, *M_w*=58,000), methanol (CH₃OH, AR), N, N-dimethylformamide (DMF, 97 %), were purchased from Macklin Chemical Reagent Co., Ltd, China. 5-octylfuran-2(5H)-one (butenolide, named as Bu) was purchased from Hubei Shishun Biotechnology Co., Ltd. Graphite powder (325 mesh, 99.8 %) was produced by Alfa Aesar Chemical Co., Ltd. Hydroxy-terminated fluorinated polysiloxane (OH-^FPDMS, 2000CS) was produced by Guangzhou Fuze New Materials Co., Ltd., China. 4,4'-methylene-bis (2-chloroaniline) (MOCA, 98 %), 1, 6-hexapropylisocyanate (HDI, 97 %) were produced by Anage Chemical Co., Ltd., China. Ethylene glycol (EG, 99.7 %), sodium chloride (NaCl, 99.5 %), and ethanol (C₂H₅OH, 99.7 %) were purchased from Tianjin Best Chemical Co., Ltd., China. Silver nitrate (AgNO₃, 99.7 %) was purchased from the Shanghai Institute of Fine Chemical Materials.

Tryptone, yeast infusion powder, and agar powder (BR) were all purchased from Beijing Oboxing Biotechnology Co., Ltd., China. *Escherichia coli*, *Staphylococcus aureus*, and *Candida albicans* were all purchased from Shanghai Luwei Strain Biotechnology Research Center, China. Chlorella and its culture medium were purchased from Nanjing Hailes Biotechnology Co., Ltd., China. Cold-rolled sandblasted steel plates (150 mm × 70 mm × 0.8 mm) were purchased from Hongtu Metal Factory, China. Primer was purchased from Shanghai Xuan Yang Chemical Materials Technology Co., Ltd., China. Polyurethane was purchased from Shaanxi Xitai Xingbang Materials Technology Co., Ltd., China.

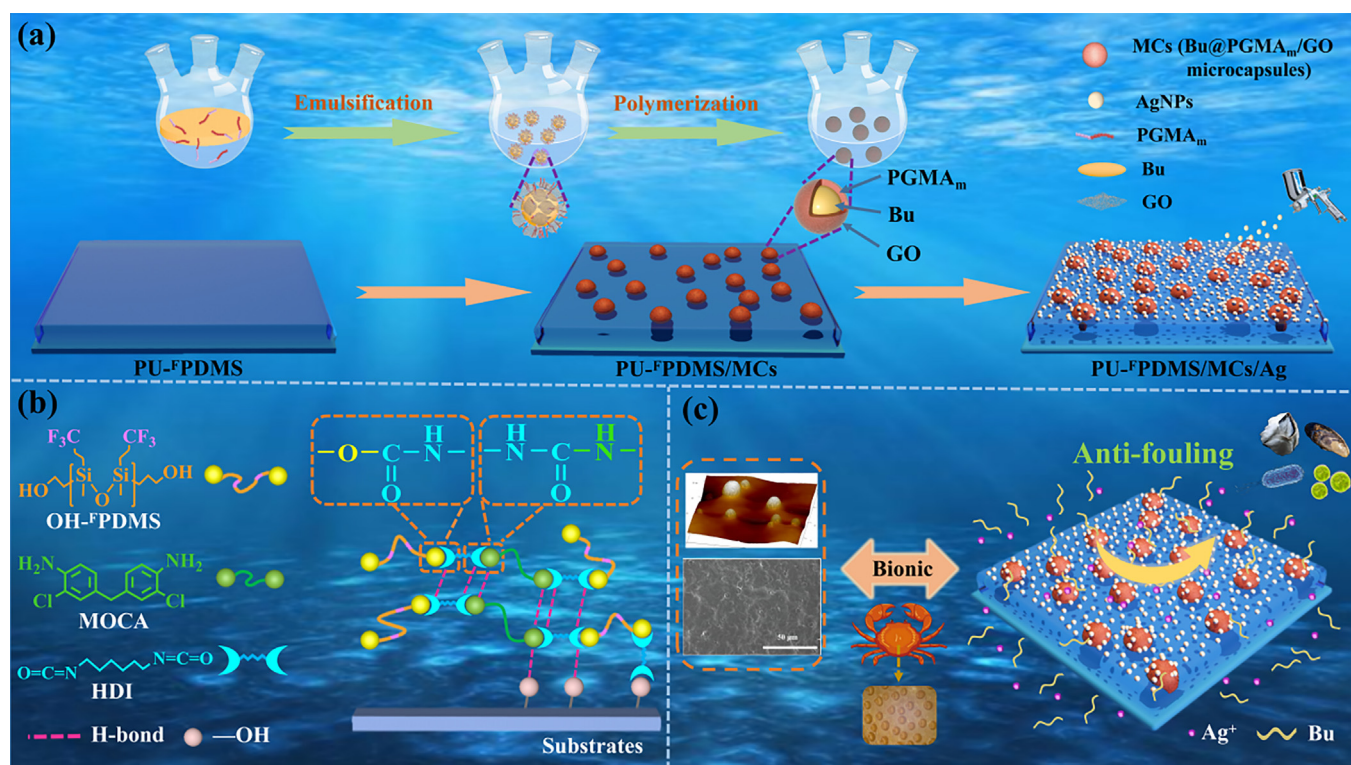


Fig. 1. (a) Preparation of PU-FPDMS/MCs/Ag bionic anti-fouling coating, (b) molecular interaction and bonding mechanism of robust PU-FPDMS matrix, and (c) multi-strategy combined anti-fouling mechanism.

2.2. Preparation of PU-FPDMS/MCs/Ag bionic anti-fouling coatings

GO was prepared using the modified Hummers method [35], and the GO aqueous solution was diluted to a concentration of 0.95 wt% for later use. 14.04 g of DPE-modified hydrolyzed polyglycidyl methacrylate [36,37] (PGMA_m, 17.80 wt%), 4.69 g of DVB, 6.00 g of Bu, 13.16 g of GO aqueous solution, and 11.98 g of H₂O, were added sequentially to a 250 mL three-necked round-bottom flask. Emulsification was conducted at 1500 r min⁻¹ speed for 10 min. Then, it was transferred to an oil bath, refluxed, and reacted at 80 °C for 10 h under 200 r min⁻¹. The product was centrifugally washed and freeze-dried to obtain Bu@PGMA_m/GO microcapsules with a core content of 73.25 wt%. The calculation method of the core content is shown in Supplementary Material.

3 g of OH-FPDMS, 0.55 g of HDI, 0.03 g of DOTDL, 1.08 g of MOCA solution (0.13 g of MOCA, acetone as solvent), and 0.15 g of Bu@PGMA_m/GO microcapsules were mixed well and degassed under vacuum. Then, the mixture was pouring into a mold or coated on different substrates. After that, 5 wt% of homemade AgNPs acetone solution was sprayed, and cured at room temperature for 72 h to obtain PU-FPDMS/MCs/Ag bionic anti-fouling coating. Among them, the preparation process of AgNPs is shown in Supplementary materials. In the above process, AgNPs were not added to obtain PU-FPDMS/MCs coating, Bu@PGMA_m/GO microcapsules were not added to obtain PU-FPDMS/Ag coating, and Bu@PGMA_m/GO microcapsules with AgNPs were not added to obtain PU-FPDMS coating. The preparation routes of Bu@PGMA_m/GO microcapsules and PU-FPDMS/MCs/Ag bionic anti-fouling coating are shown in Fig. 1.

2.3. Characterizations

The morphology of the samples was observed by a Verios G4 scanning electron microscope (FEI, USA), a metallurgical mi-

croscope 10XB-PC (Shanghai Optical Instrument Factory, China), and a Dimension Icon atomic force microscope (Bruker, Germany). The size distribution of microcapsules was measured by a laser particle analyzer DLS LS13320 (Beckman Coulter Instruments, USA). The chemical composition and structure of the samples were characterized by a Fourier transform infrared spectrometer Tensor 27 (Bruker, Germany), an X-ray diffractometer XRD-7000 (Shimadzu, Japan), a Raman spectrometer Alpha300R (WITec, Germany), an X-ray photoelectron spectroscopy Axis Supra (Shimadzu/Kratos, Japan), and a 600 MHz nuclear magnetic resonance scanner (Bruker, Germany). Thermogravimetric analyses were carried out using a TGA2 apparatus (Mettler Toledo, Switzerland), and the sample was heated from 35 to 700 °C at a heating rate of 10 °C min⁻¹ in nitrogen. The absorbance of the solution was measured by a UV-visible spectrophotometer (Cary5000 Agilent, USA).

A universal testing machine QJ211S (Shanghai Qingji Instrumentation Technology Co., Ltd., China) was used to test the mechanical properties of specimens. The samples were coated on glass blocks (30 mm × 20 mm × 15 mm) and their shear strengths were tested with a compression shear speed of 2 mm min⁻¹ according to QJ 1634A-1996. Tensile speeds of the dumbbell specimens were 50 mm min⁻¹ and the tensile test was accorded to ASTM D412 standard. The abrasion resistance test was carried out using polyethylene terephthalate (PET) fabric as the abrasion interface, and the specimen was fixed to a stainless steel column with reciprocating motion at a pressure of 45 kPa (one cycle of 20 cm), performed according to the modified AATCA test method 8-2001. The electrochemical impedance spectrums of the samples were tested using an electrochemical workstation ParStat 4000 (Princeton, USA) and a three-electrode cell, in which the coating on the copper substrate was used as the working electrode. The platinum sheet was the counter electrode, and the saturated calomel was used as the reference electrode. The electrolyte was 3.5 wt% NaCl

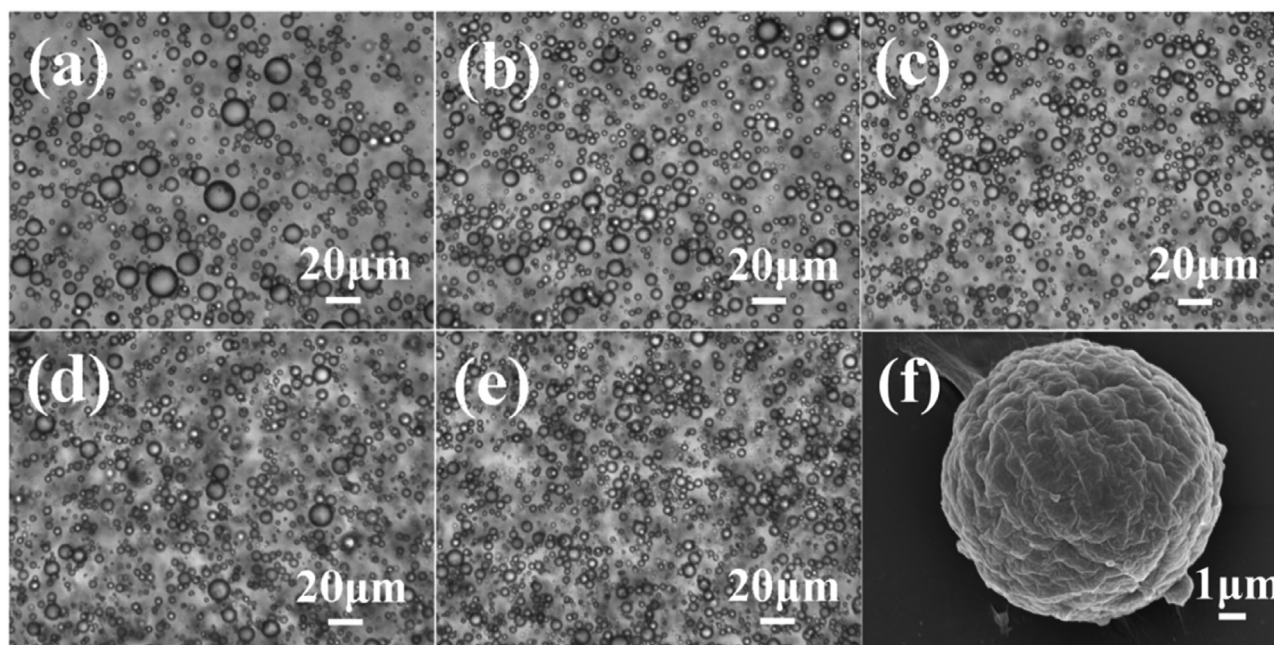


Fig. 2. OM images of (a) emulsion formed by the synergistic action of GO sheets and PGMA_m, Bu@PGMA_m/GO microcapsules formed at different reaction times of (b) 1 h, (c) 4 h, (d) 7 h, (e) 10 h, and (f) SEM images of the Bu@PGMA_m/GO microcapsules after 10 h reaction.

solution, the frequency range was 10^{-2} – 10^5 Hz for Nyquist and Bode plots, and the potential range was -0.5 to 1.5 V (vs OCP) with a scan rate of 1 mV S^{-1} for the polarization curves. The anti-bacterial performance and anti-algae attachment performance of the samples were tested using an ultra-clean workbench SW-CJ-1D (Shanghai Shangdao Instrument Manufacturing Co., Ltd., China), a shaking incubator BS-1E (Shangyu Xingxing Instrument Equipment Co., Ltd., China), and a high-pressure steam autoclave YX-280D (Hefei Huatai Medical Equipment Co., Ltd., China). The concentration of bacterial solution was tested using a bacterial turbidity meter WZG-XT (Hangzhou Qiwei Technology Co., Ltd., China). The tests of anti-bacterial performance, anti-algae attachment performance, and anti-marine biofouling attachment performance are shown in Supplementary materials.

3. Results and discussion

3.1. Morphology of Bu@PGMA_m/GO microcapsules

Fig. 2 shows the OM images of the emulsion formed by the synergistic action of GO nanosheets and PGMA_m, as well as the OM and SEM images of Bu@PGMA_m/GO microcapsules at different reaction time. As can be seen from Fig. 2(a), the water phase is clear and transparent, and GO sheets aren't be found in the water phase, indicating that the GO sheets are effectively captured by the amphiphilic macromolecule PGMA_m in the water phase under hydrogen bonding, and the two are evenly dispersed at the water-oil interface, which play a good synergistic stabilizing effect on the emulsion. After 1 h of the reaction, the particle size of the microcapsules becomes slightly smaller (Fig. 2(b)), which is attributed to the in-situ polymerization of DVB monomer in the oil phase initiated by PGMA_m at the interface to form a polymer shell. At the same time, the GO sheets at the interface are assembled and stacked along the polymer shell into a compact shell structure. As the polymerization reaction proceeds, the particle size of the microcapsules decreases further (Fig. 2(c, d)). This is because, under heating conditions, the semiquinone-type structure of DPE in PGMA_m acquires enough energy and generates macro-

molecular free radicals, which re-initiates the polymerization of the monomers in the droplets. The conversion of monomers to polymers is a volume-reduction process, resulting in a decrease in the average particle size of the microcapsules. At the end of the reaction, the average particle size of the Bu@PGMA_m/GO microcapsules is 12.21 μm (Figs. 2(e) and S2 in Supplementary materials), and the microcapsules exhibit a distinct fold morphology due to the assembly and stacking of the GO sheets on the polymer shell (Fig. 2(f)). This compact multilayer shell formed by the polymer and GO gives the Bu core excellent stability.

3.2. Chemical composition of Bu@PGMA_m/GO microcapsules

Fig. 3(a–d) shows the FT-IR, ^1H NMR, and ^{13}C NMR spectra together with the TGA curves of Bu@PGMA_m/GO microcapsules, shells, and Bu core, respectively. As shown in Fig. 3(a), in the spectra of microcapsules and shells, the broad peak at 3350 cm^{-1} corresponds to the stretching vibration of O–H, the peak at 1720 cm^{-1} corresponds to the stretching vibration of C=O, the peak at 1050 cm^{-1} corresponds to the stretching vibration of C–O in C–O–C, indicating the existence of GO in the shells. Meanwhile, the peak at 1629 cm^{-1} corresponds to the stretching vibration of C=C in the aromatic ring indicating that PGMA_m is also contained in the shells [38]. In the spectrum of the core, the peak at 3097 cm^{-1} corresponds to the stretching vibration of =CH, and the peaks at 810 , 2925 , and 2850 cm^{-1} all correspond to the stretching vibration of $-(\text{CH}_2)_n-$ ($n > 4$), which proves that the chemical structure of Bu core contains long carbon chains [39–41]. Furthermore, these characteristic peaks all appear in the spectrum of the microcapsules, indicating that the core is successfully encapsulated by the shells and maintains effectiveness.

As shown in Fig. 3(b, c), all characteristic signals of the Bu core are present in the spectra of the microcapsules, indicating that the Bu core has been successfully encapsulated and has maintained its effectiveness. As shown in Fig. 3(d), the core begins to decompose at 147°C (mass loss of 5 wt%) and is completely decomposed at 205°C (mass loss of 95 wt%). The decompositions of microcapsules and shells are divided into three steps. The first step is between

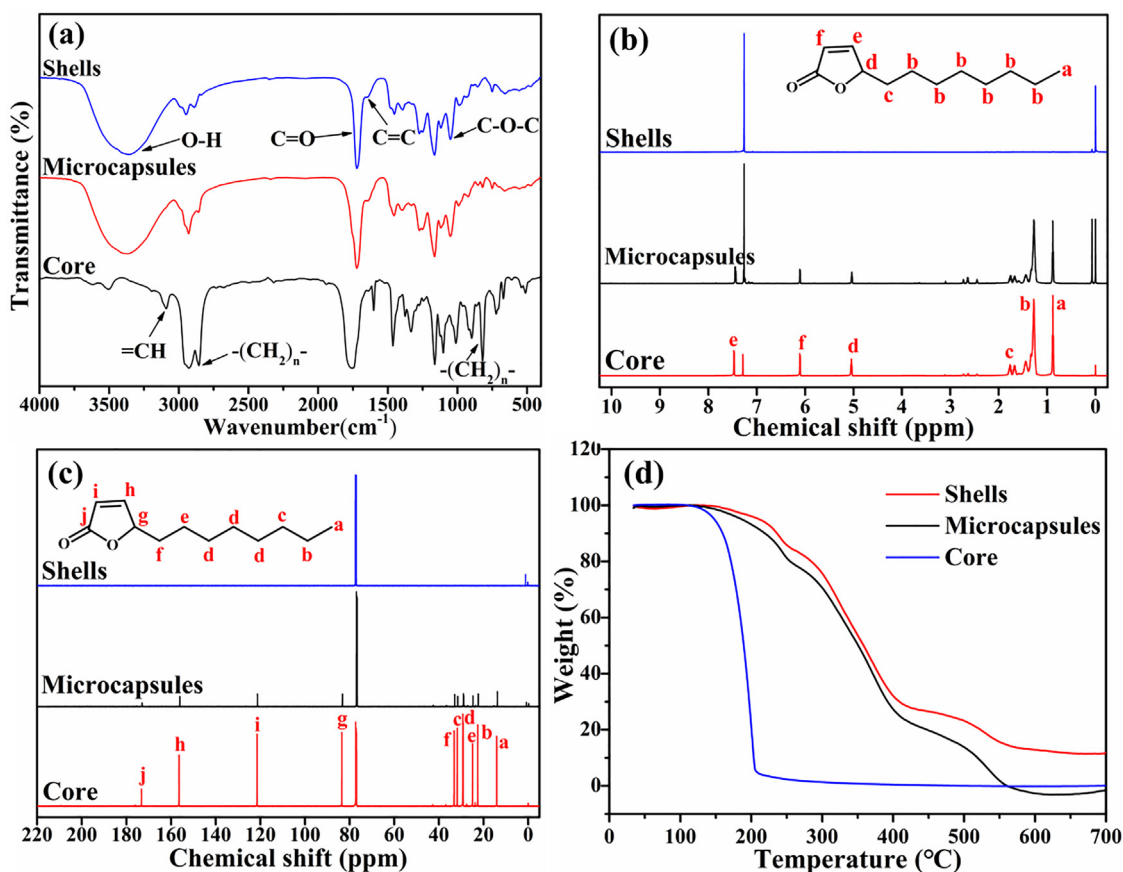


Fig. 3. (a) FT-IR spectra, (b) ¹H NMR spectra, (c) ¹³C NMR spectra, and (d) TGA curves of microcapsules, shells, core, and GO.

150 and 250 °C, which is mainly caused by the decomposition of oxygen-containing groups in the GO outer shell. The second step is between 250 and 400 °C, which is mainly caused by the decomposition of PGMA_m in the polymer inner shell. The third step is to decompose above 400 °C, mainly due to the decomposition of the shells, which leads to a large number of decomposition of the core. Because of the protection of the compact shell, the thermal stability of the core is significantly improved.

Fig. S3 shows the XRD, Raman, and XPS spectra of GO, Bu@PGMA_m/GO microcapsules and their shells. As shown in Fig. S3(a), the characteristic diffraction peaks of GO, the microcapsules, and their shells all appear at 11.84°. As shown in Fig. S3(b), the characteristic peaks of the microcapsules and shells appear at 1351 and 1591 cm⁻¹ [42], which are consistent with the D and G peaks of GO respectively, indicating the existence of GO in the microcapsules. The intensity ratio of the D peak to the G peak (I_D/I_G) represents the ratio of disordered and ordered structures of the graphene plane. The I_D/I_G values of the microcapsules and their shells are similar, and they both are smaller than those of GO. As shown in Fig. S3(c, d) and Table S1, the content of oxygen-containing groups and the O/C ratio of microcapsules are smaller than those of GO. These show that GO interacts with PGMA_m, and GO is assembled and stacked on the PGMA_m polymer shell, which reduces defects and disordered structure of GO, improves the order degree of GO, and a certain reduction of GO is obtained.

3.3. Release behavior of Bu

Fig. 4(a, b) shows the release concentration and leakage rate of Bu after the Bu@PGMA_m/GO microcapsules were soaked in

methanol for different days. As shown in Fig. 4(b), the leakage rate of Bu is 37.03 % after 5 days of immersion and 47.38 % after 10 days. From day 10 to day 70, Bu shows a slow release trend. The release behavior of the Bu core in the microcapsules is fitted by four release models: Zero-order [43], Ritger-Peppas [44], Higuchi [45], and Baker-Lonsdale [46], as shown in Fig. 4(c–f). The results show that the release behavior of Bu is most consistent with the Ritger-Peppas model. The release of Bu is related to polymer relaxation and solute diffusion, which belongs to pseudo-Fickian [47,48]. The shell relaxes, the polymer network expands, and Bu escapes from the polymer inner shell. At the same time, due to the destruction of the sp² conjugate network structure of the GO outer shell, Bu continues to pass through the GO outer shell to be released. The results indicate that the rapid release of the Bu core is limited under the action of the compact multilayer shell, and the microcapsules obtain a sustained release effect.

3.4. Mechanical properties of PU-^FPDMS/MCs/Ag

Fig. 5 shows the shear strength, tensile strength, and elongation at the break of PU-^FPDMS, PU-^FPDMS/Ag, PU-^FPDMS/MCs, and PU-^FPDMS/MCs/Ag. As shown in Fig. 5(a), the shear strengths of PU-^FPDMS, PU-^FPDMS/Ag, PU-^FPDMS/MCs, and PU-^FPDMS/MCs/Ag are 3.40, 3.42, 3.51, and 3.53 MPa, respectively. The results show that the hydrogen bonding formed by the PU-^FPDMS matrix with the substrate shows good adhesion, and the effect of filling the microcapsules and AgNPs on the shear strength is not great. As shown in Fig. 5(b), the tensile strength and elongation at the break of the dumbbell specimen of PU-^FPDMS are 6.14 MPa and 213.51 %, respectively, showing good strength and toughness. This is due to the

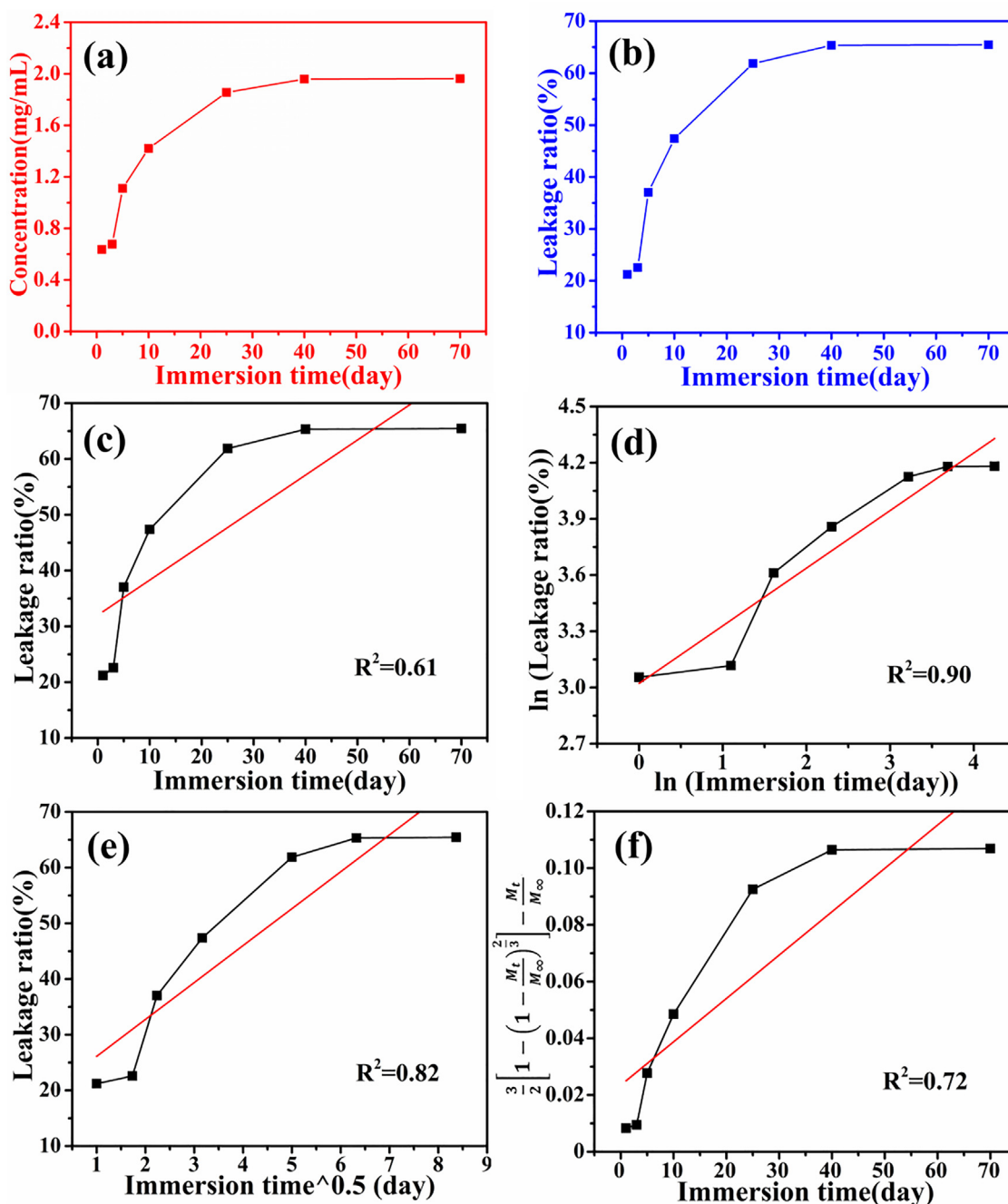


Fig. 4. (a) The release concentration, (b) leakage rate of Bu, and the fitting curves of the release behavior of Bu: (c) Zero-order model, (d) Ritger-Peppas model, (e) Higuchi model, and (f) Baker-Lonsdale model.

construction of a double cross-linking network of PU-^FPDMS under the joint action of covalent bonds and hydrogen bonds, which makes the four specimens have good mechanical strength [49]. Furthermore, the introduction of microcapsules and AgNPs plays a certain role in enhancing the PU-^FPDMS matrix, so that the tensile strength of PU-^FPDMS/Ag, PU-^FPDMS/MCs, and PU-^FPDMS/MCs/Ag reach 6.24, 6.5, and 6.7 MPa, respectively.

3.5. Self-cleaning performance of PU-^FPDMS/MCs/Ag

Fig. 6(a) shows the static contact angle and sliding angle of water droplets on PU-^FPDMS and PU-^FPDMS/MCs/Ag. As shown in Fig. 6(a), the static contact angle and sliding angle of PU-^FPDMS

are 115.6° and 67.2°, respectively, while the static contact angle and sliding angle of PU-^FPDMS/MCs/Ag are 161.8° and 3.6°, respectively. Fig. 6(b) shows the static contact angles and sliding angles of water droplets after repeated wear for PU-^FPDMS/MCs/Ag. As exhibited in Fig. 6(b), after 200 cycles of wear, the static contact angle and sliding angle become 142.0° and 13.5°, respectively, indicating that PU-^FPDMS/MCs/Ag has stable hydrophobicity. Fig. 6(c-e) shows the static contact angle and photos of contaminant droplets on PU-^FPDMS, PU-^FPDMS/MCs/Ag (coated on various substrates containing glass, ceramic, polyethylene film, cotton, and steel). As shown in Fig. 6(c-e), the static contact angles of contaminant droplets on the surfaces of PU-^FPDMS and PU-^FPDMS/MCs/Ag both are higher, and the static contact angles of contaminant droplets on the sur-

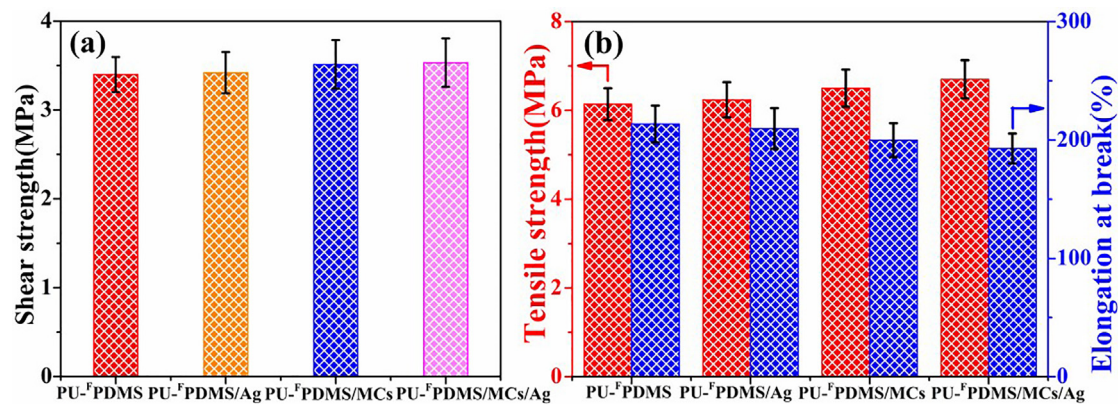


Fig. 5. (a) Shear strength and (b) tensile strength and elongation at break of PU-FPDMS, PU-FPDMS/Ag, PU-FPDMS/MCs, and PU-FPDMS/MCs/Ag.

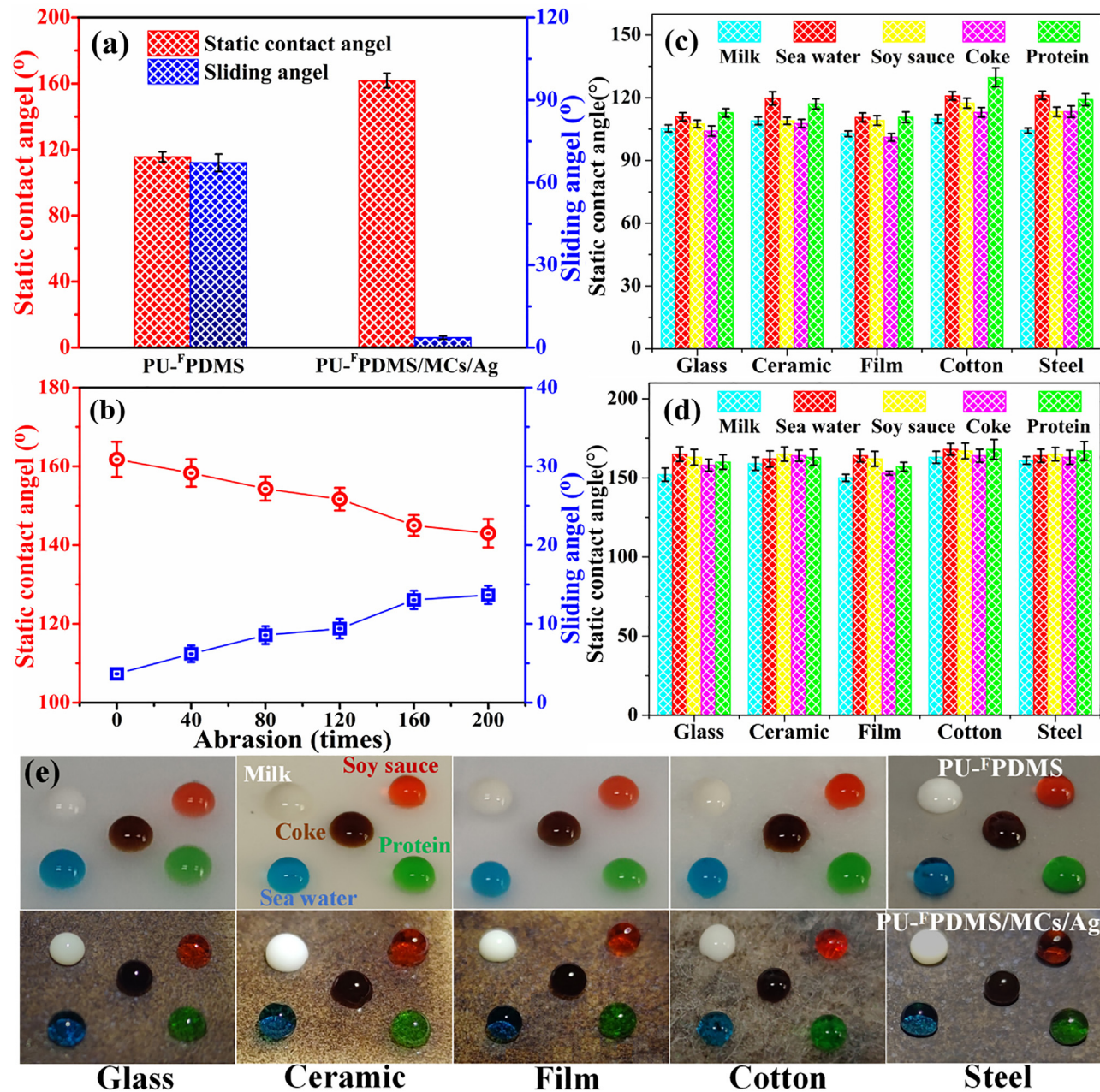


Fig. 6. (a) Static contact angle and sliding angle of water droplets on PU-FPDMS and PU-FPDMS/MCs/Ag coating, (b) static contact angle and sliding angle of water droplets after multiple wears for PU-FPDMS/MCs/Ag, and static contact angle of different contaminant droplets on (c) PU-FPDMS, (d) PU-FPDMS/MCs/Ag, and (e) photos of different contaminant droplets on PU-FPDMS and PU-FPDMS/MCs/Ag.

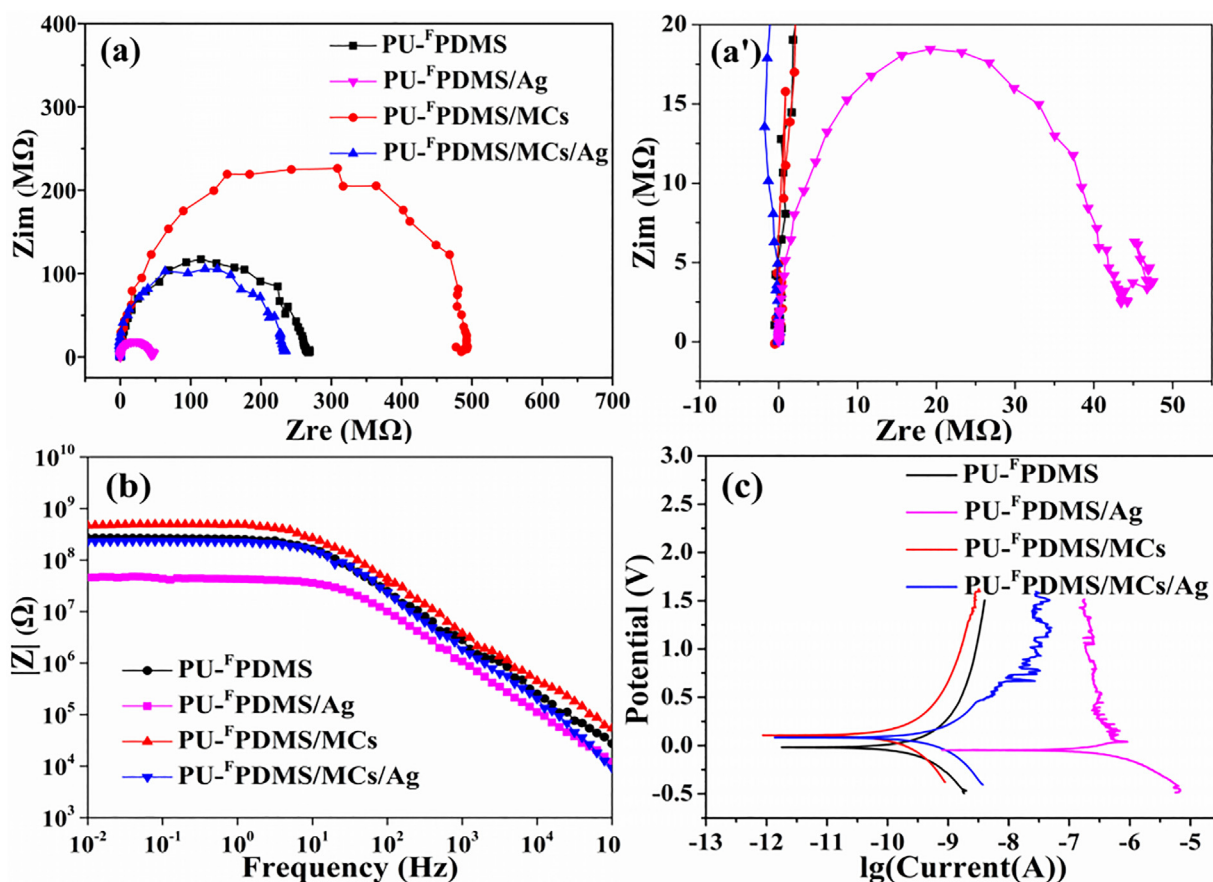


Fig. 7. (a, a') Nyquist plot, (b) Bode plot, and (c) potentiodynamic polarization curves of PU-^FPDMS, PU-^FPDMS/Ag, PU-^FPDMS/MCs, and PU-^FPDMS/MCs/Ag.

faces of PU-^FPDMS/MCs/Ag coated on different substrates are significantly higher than those on the surfaces of PU-^FPDMS/MCs/Ag. The results show that the PU-^FPDMS matrix with low surface energy and the bionic micro-nano surface formed by the microcapsules and AgNPs [34,50] (Fig. S6) play a key role in the self-cleaning performance of PU-^FPDMS/MCs/Ag.

3.6. Electrochemical performance of PU-^FPDMS/MCs/Ag

The Nyquist plot and Bode plot of PU-^FPDMS, PU-^FPDMS/Ag, PU-^FPDMS/MCs, and PU-^FPDMS/MCs/Ag are shown in Fig. 7(a, b), respectively. The equivalent circuit diagram exhibited in Fig. S7 is used to simulate the electrochemical parameters, and the relevant electrochemical parameters are shown in Table S2. As shown in Fig. 7(a, a') and Table S2, the arc radii of the four coatings all reach the order of $M\Omega$, showing good resistance to electrolytes. Compared with PU-^FPDMS, PU-^FPDMS/MCs have a larger R_{coat} value. Compared with PU-^FPDMS/Ag, the values of R_{coat} and R_{ct} for PU-^FPDMS/MCs/Ag are higher, indicating that the embedding of microcapsules greatly enhances the spatial barrier effect of the coating.

As can be seen from Fig. 7(b), in the low-frequency region (frequency $< 10^1$), the value of $|Z|$ for PU-^FPDMS/Ag is the largest, followed by PU-^FPDMS/MCs/Ag, PU-^FPDMS, the value of $|Z|$ for PU-^FPDMS/MCs is the smallest, which is consistent with the results of the Nyquist plot. The slope of $|Z|$ is related to the electrochemical response characteristic of the coating capacitance. The closer the value is to -1 , the closer the coating capacitor is to pure capacitance. When the frequency is 10^1 – 10^5 , the slopes of the $|Z|$ values of the four coatings are all close to -1 , indicating

that the corrosion rates of the coatings in the electrolyte are low [51,52].

As can be seen from Fig. 7(c) and Table S3, the U_{corr} value of PU-^FPDMS/Ag is the smallest and the I_{corr} value is the largest, indicating that after AgNPs are embedded in PU-^FPDMS, the Ag^+ dissolved in the surface of the coating participates in the electrochemical reaction, which accelerates the dissolution of Ag^+ on the surface of the coating and leads to the acceleration of the corrosion rate of the coating. Compared with PU-^FPDMS/Ag, the U_{corr} value of PU-^FPDMS/MCs/Ag significantly increases, while the I_{corr} value significantly decreases, indicating that the embedding of the microcapsules improves the spatial barrier effect of the coating. In addition, the GO shell in the microcapsule has a high surface area and resistance to permeability to water, oxygen, and ions. Thus, the negative effects caused by the electrochemical reaction of AgNPs are compensated [53], which is consistent with the results of electrochemical impedance spectroscopy (Fig. 7(a, a', b)).

3.7. Antibacterial performance of PU-^FPDMS/MCs/Ag

Fig. 8 shows the results of solvent blank, PU-^FPDMS, PU-^FPDMS/Ag, PU-^FPDMS/MCs, PU-^FPDMS/MCs/Ag against *Escherichia coli*, *Staphylococcus aureus*, and *Candida albicans*. As shown in Fig. 8(a, b), for the above three strains, there are no bacteriostatic circles in the solvent blank, and the bacteriostatic circle diameters of PU-^FPDMS/MCs/Ag are the maximum, and the bacteriostatic circle diameters of PU-^FPDMS are the minimum. As shown in Fig. 8(c), after mixing $200 \mu g mL^{-1}$ of PU-^FPDMS/MCs/Ag samples with the three bacterial solutions for 5 min, respectively, the antibacterial rates against the three kinds of bacteria reach 100 %.

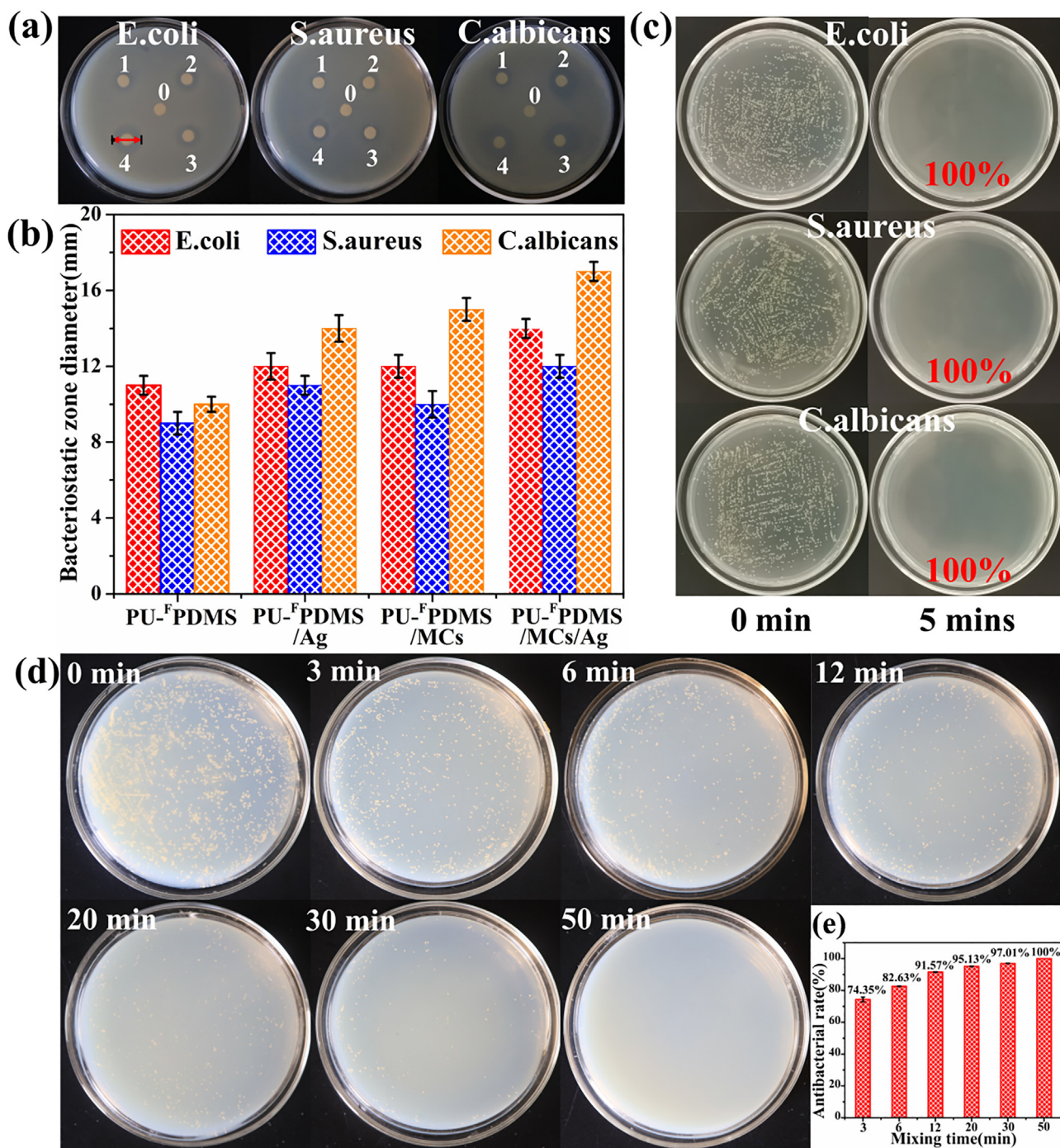


Fig. 8. (a) Photos of diffusion test of filter paper for solvent blank (0), PU-FPDMS (1), PU-FPDMS/Ag (2), PU-FPDMS/MCs (3), and PU-FPDMS/MCs/Ag (4), (b) diameter of bacteriostatic circle for the samples, (c) photos of colony count test for 200 $\mu\text{g mL}^{-1}$ of PU-FPDMS/MCs/Ag samples, and (d) photos of colony counting experiment and (e) antibacterial rate for 100 $\mu\text{g mL}^{-1}$ of PU-FPDMS/MCs/Ag samples at different times.

As shown in Fig. 8(d), 100 $\mu\text{g mL}^{-1}$ of PU-FPDMS/MCs/Ag sample is mixed with *Candida albicans* solution at different times, it is found that the number of colonies on the medium decreases with the prolongation of mixing time. After mixing for 12 min, the antibacterial rate of *Candida albicans* reaches 91.57 %. After mixing for 50 min, it reaches 100 %. These show that the PU-FPDMS matrix, AgNPs, and microcapsules all have certain antibacterial activity. Meanwhile, the three components play a synergistic antibacterial role, which is conducive to the excellent antibacterial performance of PU-FPDMS/MCs/Ag.

3.8. Anti-algal attachment performance of PU-FPDMS/MCs/Ag

Fig. 9 shows the algae attachment of glass, PU, PU-FPDMS, PU-FPDMS/Ag, PU-FPDMS/MCs, and PU-FPDMS/MCs/Ag after immersing in chlorella for 15 days. As shown in Fig. 9(a, b), after 15 days, chlorella is denser on the surfaces of glass and PU, followed by PU-FPDMS, while chlorella attachments are less obvious on the surfaces of PU-FPDMS/MCs and PU-FPDMS/MCs/Ag. As shown in Fig. 9(c, d), the number and the coverage percentage of chlorella, are glass, PU, PU-FPDMS, PU-FPDMS/Ag, PU-FPDMS/MCs, and PU-

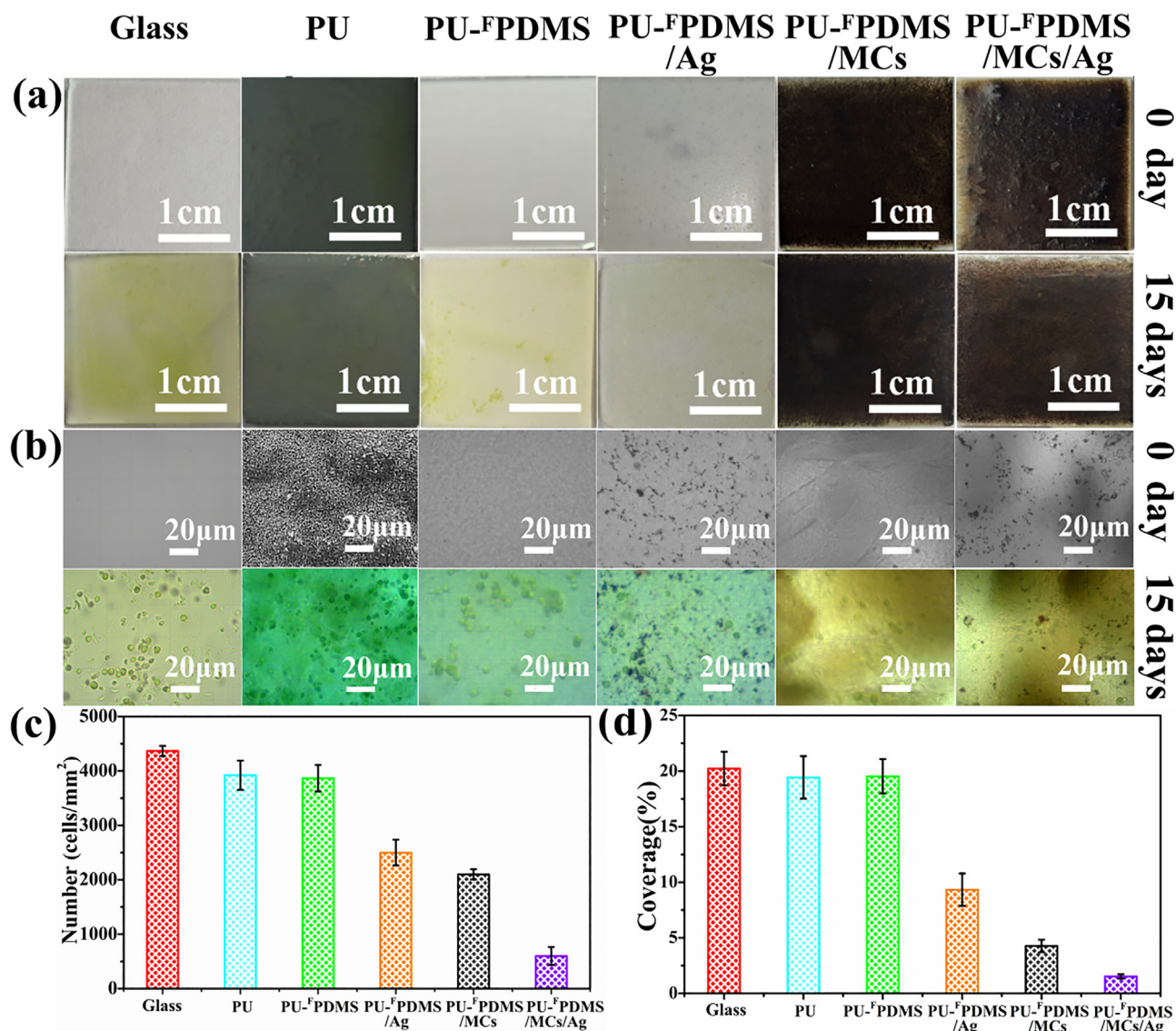


Fig. 9. Glass, PU, PU-^FPDMS, PU-^FPDMS/Ag, PU-^FPDMS/MCs, PU-^FPDMS/MCs/Ag soaked in chloroella (a) photo, (b) microscope image, (c) adhesion number, and (d) coverage rate of chloroella.

^FPDMS/MCs/Ag in order from the highest to the lowest. Among them, the number of chloroella on the surface of PU-^FPDMS/MCs/Ag is 600 cell mm⁻² and the coverage percentage is 1.53 %. The results indicate that the PU-^FPDMS coating with low surface energy has certain anti-algal attachment properties. In addition, the excellent anti-algae attachment performance of PU-^FPDMS/MCs/Ag is obtained under the multi-strategy combination of the sustained-release effect of Bu, the dissolution of Ag⁺, and the bionic micro-nano surface formed by microcapsules and AgNPs [54].

3.9. Anti-marine biofouling performance of PU-^FPDMS/MCs/Ag

Fig. 10 shows the photos of marine panels in the seawater around Qiongzhou Strait (20° 01'N, 110° 16'E) in the South China Sea. As can be seen from Fig. 10, a relatively obvious biofilm is attached to the surface of the blank coating after one month, and only sporadic biofilm appears on the surfaces of the PU and PU-^FPDMS coatings, while biofilm is not found on the surface of the PU-^FPDMS/MCs/Ag coating [55]. After 2 months, the biofilm cover-

age area of the blank and PU increased, and bryozoans and aquatic animals were clearly visible on their surfaces. The PU-^FPDMS coating shows sporadic attachment of bryozoan locally, while the PU-^FPDMS/MCs/Ag coating does not show biofouling coverage [56]. After 4 months, for the blank and PU coatings, the biofilm covers their entire surface and the thickness of the biofilm increases, meanwhile, shellfish adhesion is visible. For the PU-^FPDMS coating, the coverage area of biofilm increases, and the coverage area of bryozoan increases, while the PU-^FPDMS/MCs/Ag coating still does not show biofouling coverage. After 6 months, large shellfish, spirorbis, and algae could be attached to the blank coating, algae and larvae of spirorbis could be seen on the surface of the PU coating, a small amount of algae and larvae of spirorbis are attached to the PU-^FPDMS coating, and the PU-^FPDMS/MCs/Ag coating is still not covered with biofouling. These results show that the PU-^FPDMS/MCs/Ag has excellent anti-fouling performance in marine organisms under the combined action of PU-^FPDMS matrix with low surface energy, sustained release of Bu, dissolution of Ag⁺, and bionic surface formed by the microcapsules and AgNPs.

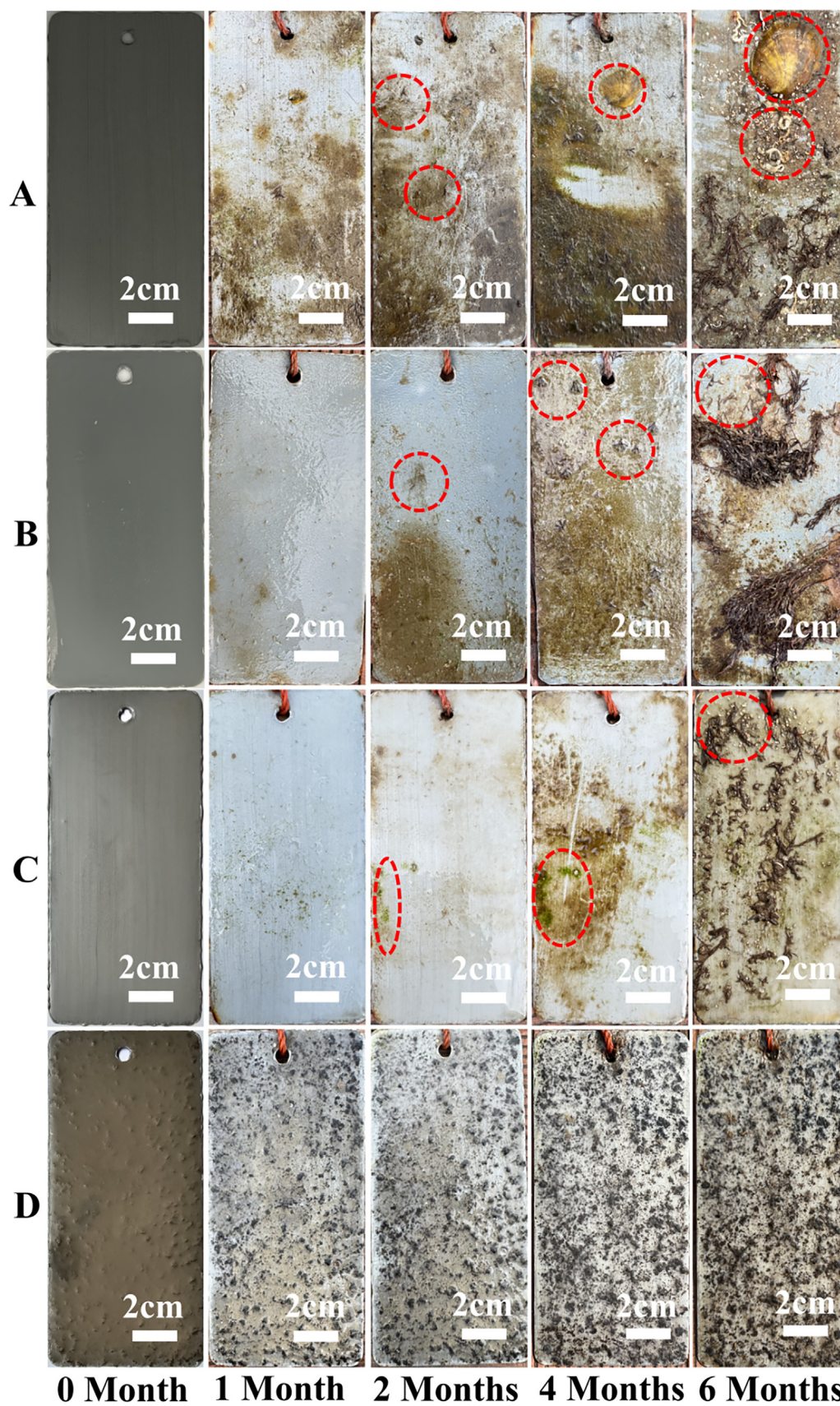


Fig. 10. Photos of marine panels completed in the seawater near the Qiongzhou Strait (20° 01'N, 110° 16'E) in the South China Sea. (A) The panels with the prime coating (Blank); (B) the panels with PU coating on the prime coating; (C) the panels with PU-FPDMS coating on the prime coating; (D) the panels with PU-FPDMS/MCs/Ag coating on the prime coating.

4. Conclusions

Bu@PGMA_m/GO microcapsules with compact multi-shell structures have been successfully prepared. The PU-^FPDMS/MCs/Ag bionic anti-fouling coating with a crab shell-like surface is obtained by assembling the Bu@PGMA_m/GO microcapsules and Ag-NPs step by step in the PU-^FPDMS matrix with high strength and high bonding. The release behavior of Bu conforms to the Ritger-Peppas model and is related to polymer relaxation and solute diffusion, which belongs to pseudo-Fickian diffusion. The shear strength, tensile strength, and elongation at break of PU-^FPDMS/MCs/Ag are 3.53 MPa, 6.7 MPa, and 192.83 %, respectively. The static contact angle and sliding angle of water on the PU-^FPDMS/MCs/Ag coating are 161.8° and 3.6°, respectively. The antibacterial rate of PU-^FPDMS/MCs/Ag against *Escherichia coli*, *Staphylococcus aureus*, and *Candida albicans* can reach 100 %. After immersing in chlorella for 15 days, the adhesion number and coverage percentage for PU-^FPDMS/MCs/Ag are the smallest, which are 600 cell mm⁻² and 1.53 %, respectively. After 6 months of marine field tests, the PU-^FPDMS/MCs/Ag coating is still not covered with biofouling. PU-^FPDMS/MCs/Ag has the excellent anti-fouling performance of marine organisms under the combined action of a robust polymer matrix with low surface energy, steady-state sustained release of Bu encapsulated by compact multilayer shell, dissolution of Ag⁺, and bionic micro-nano surface formed by microcapsules and AgNPs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Nan Zheng: Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization. **Bo Jia:** Formal analysis, Data curation. **Jie Liu:** Writing – review & editing, Supervision, Conceptualization. **Xiaojun Wang:** Software, Methodology. **Duo Zhang:** Formal analysis. **Hairan Zhang:** Methodology. **Guoqing Wang:** Supervision, Project administration, Funding acquisition.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jmst.2024.05.025](https://doi.org/10.1016/j.jmst.2024.05.025).

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